

1. High-level flow (both desks)

- Product: MIS intraday (same-day square-off). Modes: `curated_list` | `agent_auto` | `dual_desk`.
- 09:00 IST ? Pre-market analysis (no orders).
- 09:15 IST ? Market open; scheduler cycles begin.
- Each cycle: (1) Monitor exits (`position_guard` + LTP guard), (2) Check session/global caps,
- (3) Rank & pick candidates, (4) Run buy gates sequentially, (5) Execute MIS market buy,
- (6) Log pick + RAG lessons on close.
- 15:20 IST ? Mandatory EOD square-off window. 15:30 IST ? Market close; auto-stop if flat.

2. Shared scoring (`_score_screen_row`)

- Calibration base score ? historical win-rate from `calibration.json`.
- Composite stance: +12 strong_favorable, +10 favorable, +9 bullish, +7 constructive; -15 bearish.
- RSI 14: +3 if 45-65; -5 if ≥ 78 .
- Quality: +4 if `quality_ok`; -8 if `quality_fail`. Pattern: +5 bullish; -6 bearish.
- Quality score +3 if ≥ 65 . Psychology mood/sentiment adjustments.
- RAG adjustment: +/- per win/loss lesson. Symbol win-rate from `pick_log`.
- Minimum composite to trade: $\text{entry_min_composite} / \text{agent_min_composite} = 55$ (default).

3. Curated List ? stock selection

- Universe: Your list only ? 1-20 symbols (`curated_symbols` / watchlist).
- Pre-market: `learn_timing_batch()` on all curated symbols -> timing profiles + Chroma RAG.
- Data: Morning scan rows; missing symbols get live `_analyze_symbol()`.
- Ranking: `allocate_budget_picks(agent_smart=False)` ? sort by expected profit INR within budget.
- Filter: Bullish stance + composite ≥ 55 ; skip already-held symbols.
- No agent-only filters: no cap diversification, no cross-desk block, no min-notional skip.

4. Auto Pick ? stock selection

- Universe: Nifty 500 via morning scan (preferred) or screener cache.
- Pre-market: `run_morning_scan()` + `agent_premarket_learn()` ? timing on top 30 + curated overlap.
- Deep analyze: Top 20 rows missing ATR/horizon get full `_analyze_symbol()`.
- Scan depth: Top 80 rows (`agent_scan_top_n`). Ranking: `rank_rows_for_agent()` intraday edge.
- Intraday edge: composite $\times 0.22$, ATR (+18 if $\geq 2\%$), horizon return, RSI, cap bucket weights
- (small 1.25, mid 1.12, large 0.88), pattern/supertrend/golden cross, `rvol`, realistic move.
- Agent learning layer: curated overlap +18, timing profile +/-8/6, learning curve from

- RAG lessons +/-12. Diversification: max 2 large, 5 mid caps.
- Auto-only skips: cross-desk duplicate, notional < 25000 INR, agent_candidate_gate, 10:30 cutoff.

5. Agent candidate gate (auto only)

- agent_min_atr_pct: 2.0 ? block low volatility.
- agent_max_rsi_entry: 72 ? block extended RSI.
- agent_min_horizon_return_pct: 0.5 ? block negative horizon.
- agent_no_buy_after_minute_ist: 630 (10:30 IST) ? no new auto picks after this.
- agent_block_quality_fail: optional quality gate.

6. Position sizing (both desks)

- ATR risk sizing (default): qty sized so stop-out risks ~ risk_per_trade_inr.
- Effective stop = max(configured stop%, 50% of ATR%, 0.5%).
- Conviction multiplier when edge >= 75: x1.5 conservative / x2.5 home run.
- Capped by max_position_inr, remaining session budget, max_concurrent_picks slots.
- Conservative profile: risk_per_trade_inr = 3500. Home run: 8000.
- Conservative max_position_inr = 300000. Home run: 500000.

7. BUY gates ? exact order (first failure = skip)

- 1. Late session block ? no buys after 14:30 IST (no_buy_after_minute_ist=870). Both desks.
- 2. Cross-desk duplicate ? auto only; skip if other desk holds symbol.
- 3. Agent candidate gate ? auto only; ATR, RSI, horizon, 10:30 cutoff.
- 4. Stop cooldown ? no same-day re-buy after stop (stop_cooldown_enabled).
- 5. Same-day symbol block ? no re-buy after any exit today.
- 6. RAG repeat-loser ? block if >=2 RAG loss lessons OR pick_log WR <=35% with >=2 losses.
- 7. Timing gate ? preferred windows only: open_drive (9:15-9:45), morning_trend (9:45-10:30).
- 8. Latency compensation ? block negative momentum; optional early entry on pattern.
- 9. Psychology buy gate ? FOMO, euphoria, panic, defensive+bearish blocks.
- 10. Risk validate_order ? orders, loss cap, profit cap, positions, notional.
- 11. validate_auto_entry ? composite >= 55 for auto sources.
- Pre-market 9:00-9:15: plan only, no buys. Top-up: +0.5% after 10 min, max 25-50% of cap.

8. SELL exits ? priority order

- 1. eod_square_off ? >= 15:20 IST (square_off_minute_ist=920) or market closed.
- 2. stop_hit ? return <= -stop_pct.
- 3. partial_scale_out ? return >= scale-out level; sell 50% qty once per position.
- 4. trailing_stop ? peak >= trail_arm then pullback >= trail_pct from peak.
- 5. power_hour_profit_lock ? after 15:00, peak >= 0.4%, pullback >= 0.25%.
- 6. target_hit ? return >= effective target.
- 7. unrealized_loss_cap ? position loss exceeds max_unrealized_loss_inr; halts trading.
- 8. psych_take_profit ? greed + RSI>=70 + return >= 0.9%.
- 9. psych_fear_cut ? fear/defensive + return <= -0.55%.
- LTP guard: 3-second debounced tick triggers fast stop/target checks.

9. Time-decay exit parameters

- Before 14:30: base target (profile), trail_arm 1.0%, trail 0.5%.
- 14:30-15:00 power hour: target min(base, 1.0%), trail_arm 0.6%, trail 0.35%.
- After 15:00: target min(base, 0.8%), trail_arm 0.5%, trail 0.3%.
- Dynamic targets: small cap + ATR $\geq$ 3.5% -> home_run_target (3.5% conservative / 5% home run).
- ATR $\geq$ 3.0% -> extended_target (2.5% / 3.5%). Default target_pct = 2.0%.
- Partial scale-out: enabled, 50% of qty at 50% of target (~1% if target is 2%).

10. RAG and learning curve

- Trade outcomes written on every sell; read during ranking and RAG buy gate.
- Timing profiles from pre-market learn; operational lessons (Aug 14 rules).
- Session digests at session end. Morning scan chunks at 9:00. Pick log win rate accumulated.
- RAG blocks buy: ≥ 2 loss lessons OR pick_log ≥ 2 losses with WR $\leq 35\%$.
- On close: lessons feed Chroma -> tomorrow's picks rank higher/lower automatically.
- Agent also uses: timing_profile boost, curated overlap, learning_curve_adjustment.

11. Global risk caps

- max_orders_per_day: 5 default (30 in active session).
- max_intraday_loss_inr: 5000 ? halt all trading.
- max_profit_inr: 8000 ? block new buys when hit (halt_new_buys_on_profit).
- max_unrealized_loss_inr: 3000 ? halt + force exit.
- max_position_inr: 50000 default (300k conservative / 500k home run).
- max_daily_notional_inr: 100000. max_open_positions: 3.
- agent_min_composite: 55. default_stop_pct: 0.75%. default_target_pct: 1.5%.

12. Per-session caps (dual desk typical)

- max_daily_notional_inr: up to 10L per desk.
- max_profit_inr: 50k-150k per desk (profit profile).
- max_intraday_loss_inr: 15k per desk. max_concurrent_picks: 3-5 per desk.
- Session ends: desk profit cap, desk loss cap, global halt, EOD auto-stop, user stop.
- On stop: desk positions flattened (session_stop_square).
- Before restart: orphan positions auto-squared (pre_restart_square).

13. Profit profiles

- CONSERVATIVE: max_position 300k, risk/trade 3500, home_run_target 3.5%, conviction x1.5, pyramid 25%, desk profit cap 50k each. Tagline: 15-30k typical days.
- HOME RUN: max_position 500k, risk/trade 8000, home_run_target 5%, conviction x2.5, pyramid 50%, desk profit cap 150k each. Tagline: 150k ceiling on trend days.
- Auto home-run: morning score ≥ 65 switches agent to home_run; reverts if score ≤ 45 .

14. Timing windows (IST)

- pre_open 9:00-9:15 ? No trades; analysis only.
- open_drive 9:15-9:45 ? YES preferred buy window.
- morning_trend 9:45-10:30 ? YES preferred buy window.
- midday 10:30-14:30 ? Blocked for new buys if timing_preferred_windows_only=true.

- closing 15:20-15:30 ? EOD square-off. Agent hard cutoff: 10:30 for new auto picks.

15. Psychology parameters

- psychology_block_rsi_fomo: 78 ? block buy if greed + high RSI.
- psychology_block_sentiment_high: 85 ? block euphoria buys.
- psychology_block_sentiment_low: 22 ? block panic (unless fear dip + bullish).
- psychology_early_take_pct: 0.9% ? early take in greed on sell.
- psychology_fear_stop_pct: 0.55% ? tight fear cut on sell.
- psychology_greed_rsi_take: 70 ? take profit trigger.

16. Dual desk and operational guards

- Cross-desk dedup: Auto won't buy what curated already holds.
- Square on stop: stopping a desk flattens its open MIS legs.
- Square before restart: legacy orphan positions auto-squared before new session.
- Orphan banner in UI + manual Square all API endpoint.
- Mac sleep guard during session. LTP stream 3s poll for fast exits.

17. Curated vs Auto ? side by side

- Universe: Curated 1-20 your picks | Auto Nifty 500.
- Pre-market: Curated timing on all symbols | Auto scan 500 + timing top 30.
- Ranking: Curated expected profit INR | Auto edge + RAG + learning + diversification.
- Extra filters: Curated none | Auto ATR/RSI/horizon, 10:30 cutoff, min 25k, cross-desk.
- Buy gates after pick: IDENTICAL 11-step pipeline.
- Sell rules: IDENTICAL.
- RAG loop: IDENTICAL; Auto has richer ranking RAG.

18. Full autopilot config keys

- Scan: morning_scan_enabled, morning_scan_hour_ist=9, morning_scan_rag_top=60, agent_scan_top_n=80.
- Agent selection: agent_smart_selection, agent_min_atr_pct=2, agent_max_rsi_entry=72,
- agent_min_rsi_entry=35, agent_min_horizon_return_pct=0.5, agent_no_buy_after=630,
- agent_max_large_picks=2, agent_max_mid_picks=5, cap weights small/mid/large.
- Agent learning: agent_learning_enabled, agent_timing_learn_top_n=30, agent_deep_analyze_top_n=20,
- agent_rag_top_k=5, agent_curated_overlap_boost=18, min_trade_notional_inr=25000.
- Entry/exit: target_pct, stop_pct, trailing_stop_*, no_buy_after=870, square_off=920,
- partial_scale_out_*, dynamic_target_*, home_run_target_pct, extended_target_pct.
- Sizing: atr_risk_sizing_enabled, risk_per_trade_inr, conviction_*, momentum_pyramid_*, top_up_*.
- Timing: timing_intel_enabled, timing_preferred_windows_only, timing_preferred_window_ids,
- timing_gate_auto_trades, timing_strict_symbol, timing_adaptive_loosen.
- RAG: rag_block_repeat_losers, rag_block_min_loss_lessons=2, rag_block_min_pick_losses=2,
- stop_cooldown_enabled, same_day_symbol_block.
- LTP: ltp_guard_enabled, ltp_guard_debounce_seconds=3, ltp_poll_seconds=3.
- Profiles: profit_profile, auto_home_run_enabled, auto_home_run_min_score=65.
- Session: max_concurrent_picks, session_auto_stop_at_close, poll_seconds=120.

Disclaimer

This document describes the ScanBhav / Stock Analyzer autopilot system behaviour as implemented in software. Paper trading uses stub/simulated fills unless live mode is armed. Not investment advice. Verify all parameters in config.json and the Autopilot Desk UI before trading.

Educational research export ? not investment advice.